

Online PAC Labeling from a Linear Programming Perspective

Wanyu Zhang Lutong Hao Daohong Tu

Recap: AI-assisted labeling

We have a pool of unlabeled items $X_1, \dots, X_T$. Need to collect their true labels $Y_1, \dots, Y_T$.

Recap: AI-assisted labeling

We have a pool of unlabeled items $X_1, \dots, X_T$. Need to collect their true labels $Y_1, \dots, Y_T$.

Two ways to get a label $\tilde{Y}_t$:

- **ML models** $\hat{Y}_t^{(a)}$ — sometimes *wrong*, but *cheap*
- **Expert** Y_t — *correct*, but *expensive*.

Recap: AI-assisted labeling

We have a pool of unlabeled items $X_1, \dots, X_T$. Need to collect their true labels $Y_1, \dots, Y_T$.

Two ways to get a label $\tilde{Y}_t$:

- **ML models** $\hat{Y}_t^{(a)}$ — sometimes *wrong*, but *cheap*
- **Expert** Y_t — *correct*, but *expensive*.

Two Goals:

- Save cost by reducing the number of expert queries.
- Probably Approximately Correct (PAC) guarantee:

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon \quad w.p. \geq 1 - \alpha.$$

Problem setup

- For each item X_t , decide how to label it: use the **expert** or one of k **cheap models**.
- Choose action a_t from an action set $\mathcal{A} = \{E, 1, \dots, k\}$.
- Label the item with $\tilde{Y}_t = Y_t$ if $a_t = E$, else $\hat{Y}_t^{(a_t)}$.

Problem setup

- For each item X_t , decide how to label it: use the **expert** or one of k **cheap models**.
- Choose action a_t from an action set $\mathcal{A} = \{E, 1, \dots, k\}$.
- Label the item with $\tilde{Y}_t = Y_t$ if $a_t = E$, else $\hat{Y}_t^{(a_t)}$.
- For item t and action a , a labeling **loss** and a **cost**:

$$h_t(a) = \begin{cases} 0, & a = E \\ \ell(Y_t, \hat{Y}_t^{(a)}), & a \in [k] \end{cases} \quad c_t(a) = \begin{cases} 1, & a = E \\ 0, & a \in [k] \end{cases}$$

Problem setup

- For each item X_t , decide how to label it: use the **expert** or one of k **cheap models**.
- Choose action a_t from an action set $\mathcal{A} = \{E, 1, \dots, k\}$.
- Label the item with $\tilde{Y}_t = Y_t$ if $a_t = E$, else $\hat{Y}_t^{(a_t)}$.
- For item t and action a , a labeling **loss** and a **cost**:

$$h_t(a) = \begin{cases} 0, & a = E \\ \ell(Y_t, \hat{Y}_t^{(a)}), & a \in [k] \end{cases} \quad c_t(a) = \begin{cases} 1, & a = E \\ 0, & a \in [k] \end{cases}$$

Two goals: minimize *cost*, control the labeling *error*.

The cost-aware labeling LP

Let $x_{t,a} \in [0, 1]$ be the probability of taking action a on item t .

$$\begin{aligned} \min_x \quad & \underbrace{\frac{1}{T} \sum_t \sum_a c_t(a) x_{t,a}}_{\text{average cost}} \\ \text{s.t.} \quad & \underbrace{\frac{1}{T} \sum_t \sum_a h_t(a) x_{t,a}}_{\text{average error budget}} \leq \varepsilon, \quad \sum_a x_{t,a} = 1, \quad x_{t,a} \geq 0. \end{aligned}$$

The cost-aware labeling LP

Let $x_{t,a} \in [0, 1]$ be the probability of taking action a on item t .

$$\begin{aligned} \min_x \quad & \underbrace{\frac{1}{T} \sum_t \sum_a c_t(a) x_{t,a}}_{\text{average cost}} \\ \text{s.t.} \quad & \underbrace{\frac{1}{T} \sum_t \sum_a h_t(a) x_{t,a}}_{\text{average error budget}} \leq \varepsilon, \quad \sum_a x_{t,a} = 1, \quad x_{t,a} \geq 0. \end{aligned}$$

- Models multi-sources setting and item-specific costs.
- If true losses $h_t(a)$ were known, this LP is the *optimal* policy.

Dual problem and shadow price

One Lagrange multiplier $\lambda \geq 0$ for the error constraint. The dual:

$$\max_{\lambda \geq 0} -\lambda \varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Dual problem and shadow price

One Lagrange multiplier $\lambda \geq 0$ for the error constraint. The dual:

$$\max_{\lambda \geq 0} -\lambda \varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Simple decision rule from the shadow price

Given a fixed-price λ , the corresponding primal decision is given by:

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Dual problem and shadow price

One Lagrange multiplier $\lambda \geq 0$ for the error constraint. The dual:

$$\max_{\lambda \geq 0} -\lambda \varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Simple decision rule from the shadow price

Given a fixed-price λ , the corresponding primal decision is given by:

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

- Since $h_t(a)$ is unknown, we can replace it with an estimate $\hat{h}_t(a)$.
- The PAC guarantee does not require $\hat{h}_t(a)$ to be accurate, this affects the labeling cost.

Price gives a threshold policy

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Since $c_t(j) = 0$ and $c_t(E) = 1$ for $j \in [k]$, the decision rule is equivalent to:

1. $j_t^* \in \arg \min_{j \in [k]} \hat{h}_t(j), \quad s_t = \min_j \hat{h}_t(j)$ (the least uncertainty score).
2. $s_t \geq 1/\lambda \Rightarrow$ defer to expert $\quad s_t < 1/\lambda \Rightarrow$ route to model j_t^* .

Price gives a threshold policy

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Since $c_t(j) = 0$ and $c_t(E) = 1$ for $j \in [k]$, the decision rule is equivalent to:

1. $j_t^* \in \arg \min_{j \in [k]} \hat{h}_t(j), \quad s_t = \min_j \hat{h}_t(j)$ (the least uncertainty score).
2. $s_t \geq 1/\lambda \Rightarrow$ defer to expert $s_t < 1/\lambda \Rightarrow$ route to model j_t^* .

- Single source setting: recovers the original **PAC-labeling** threshold rule.
- Multi-source setting: use least uncertain model as the routing rule.

Given a fixed-price λ , the cost-aware labeling LP gives a simple threshold policy.

Given a fixed-price λ , the cost-aware labeling LP gives a simple threshold policy.

Since we use proxy $\hat{h}$, how to select λ such that we have PAC guarantees?

Given a fixed-price λ , the cost-aware labeling LP gives a simple **threshold policy**.

Since we use proxy $\hat{h}$, how to select λ such that we have PAC guarantees?

- **Offline** setting: fixed pool of T items, estimate $\hat{\lambda}$ from a calibration set.
- **Online** setting: items arrive sequentially, we update price λ_t online.

Part I — Offline PAC Labeling

Certify one price on a calibration set, then label the whole pool.

Offline: certify a price from a small labeled sample

Fixed-price rule. If true losses were known, then once λ is fixed, the decision separates across items:

$$a_t^\lambda = \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

- But accessing $h_t(a)$ requires the true label Y_t .
- So we form a plug-in policy family using fixed proxy losses:

$$a_t^\lambda = \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda \hat{h}_t(a)\}.$$

- The remaining question is how to choose λ .

Calibration idea

Label a small random subset from the whole pool and choose $\hat{\lambda}$ to minimize cost while keeping error $\leq \varepsilon$ with high probability.

Offline: compute the certified price

- 1 Draw calibration indices $I_1, \dots, I_m$ from the full pool and query their labels.
- 2 Estimate the error of every candidate price:

$$\hat{L}_m(\lambda) = \frac{1}{m} \sum_{s=1}^m h_{I_s}(a_{I_s}^\lambda), \quad U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}.$$

- 3 Choose the first price whose upper bound is below the target:

$$\hat{\lambda} = \min\{\lambda \geq 0 : U_m(\lambda; \alpha) \leq \varepsilon\}.$$

Why this is enough

In the equal-cost rule, increasing λ only moves items $AI \rightarrow \text{expert}$, so the policies are nested and $L_T(\lambda)$ is monotone. The next slide states the resulting PAC guarantee for the released labels.

Theorem (offline monotone-threshold PAC)

With probability at least $1 - \alpha$ over the calibration draw, the released labels satisfy

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon.$$

- Distribution-free, finite-sample. Falls back to expert-only if no finite price qualifies.
- Proxies affect *cost*, not *validity*: a bad proxy just defers more often.

Offline experiments: LP rule matches original PAC

One cheap source ($\varepsilon = 0.10$):

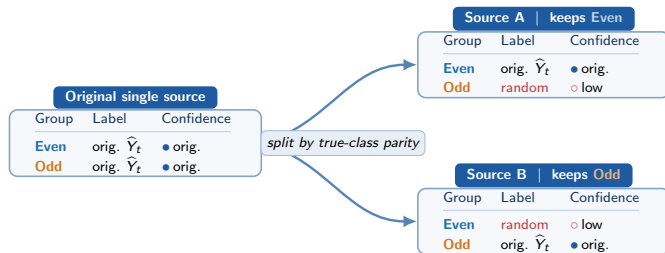
Dataset	Method	Error	Budget saved
ImageNet	LP fixed price	0.087	0.785
ImageNet	Original PAC	0.088	0.786
ImageNetV2	LP fixed price	0.072	0.546
ImageNetV2	Original PAC	0.073	0.548

- Error stays under target ε in all reported runs.
- With one cheap source, the LP fixed-price rule recovers the original PAC threshold behavior.
- On ImageNet, both methods save about 78% of expert budget.

Offline experiments: routing beats single sources

Two cheap sources ($\varepsilon = 0.10$):

Dataset	Method	Error	Budget saved
ImageNet	LP routing	0.087	0.785
ImageNet	Best single source	0.088	0.475
ImageNetV2	LP routing	0.072	0.546
ImageNetV2	Best single source	0.072	0.359



random $\sim \text{Uniform}(0, 1)$, low confidence $\sim \text{Uniform}(0, 0.2)$.

Part II — Online PAC Labeling via Audits

Items stream in; labels are hidden unless we audit.

Online PAC: moving threshold with randomized audits

Items arrive one at a time. An AI label's true loss is *hidden* unless we pay the expert.

Maintain a **moving price** λ_t , i.e. a threshold $1/\lambda_t$. Each round, route to the best model ($s_t = \min_j \hat{h}_t(j)$), then:

- $s_t \geq 1/\lambda_t$: **defer** $\rightarrow$ expert label (correct).
- $s_t < 1/\lambda_t$: release the AI label, then **audit** it with probability p .

Audit

With probability p : query the expert, *correct* the released label, and *reveal* the AI's true loss ℓ_t .

Audits buy occasional, randomized feedback — just enough to steer the price.

The price update: learn from audited feedback

Dual ascent on the price

$$\lambda_{t+1} = [\lambda_t + \eta \hat{Z}_t]_+, \quad \hat{Z}_t = \begin{cases} -\varepsilon, & \text{deferred to expert,} \\ (\ell_t - \varepsilon)/p, & \text{AI label *audited*,} \\ 0, & \text{AI label, not audited.} \end{cases}$$

- Audited **mistake** ($\ell_t > \varepsilon$) $\Rightarrow \hat{Z}_t > 0 \Rightarrow \lambda \uparrow \Rightarrow$ threshold $1/\lambda \downarrow \Rightarrow$ defer more.
- Deferral or clean audit $\Rightarrow \lambda \downarrow \Rightarrow$ trust the AI more.
- The $1/p$ re-weighting makes rare audits count correctly: on average the update sees the true AI error, $\mathbb{E}[\hat{Z}_t \mid \text{past}] = (\text{AI error before audit}) - \varepsilon$.

η : step size | ε : target error (the update centers on it directly).

Theorem (audited online control)

For *any* data sequence, with probability $\geq 1 - \delta$ over the audit draws,

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + \underbrace{\frac{\Lambda_{\text{on}}}{\eta T}}_{\substack{\text{optimization} \\ O(1/T)}} + \underbrace{\frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}}_{\substack{\text{partial feedback} \\ O(1/\sqrt{T})}} = \varepsilon + B_T.$$

- $\Lambda_{\text{on}}/(\eta T)$: how far the price can drift; vanishes like $1/T$.
- $\frac{1}{p_{\min}} \sqrt{\dots}$: noise from auditing only a fraction; smaller audit rate $\Rightarrow$ larger.
- $B_T = O(T^{-1/2}) \rightarrow 0$, so the released error approaches ε as the stream grows.

Online ($\varepsilon = 0.10$):

Data	p	Error	Saved
IN	0.1	0.078	0.663
IN	0.2	0.074	0.612
IN	0.5	0.049	0.395
INv2	0.1	0.086	0.509
INv2	0.2	0.078	0.468
INv2	0.5	0.050	0.301

- Smaller p : save more budget.
- Larger p : correct more, lower final error.

Takeaways

One idea, two regimes

- LP dual $\Rightarrow$ a single *price of error* λ .
- **Offline:** certify $\hat{\lambda}$ on held-out labels.
- **Online:** move λ_t with audited inverse-probability feedback.

What we showed

- Proxies $\hat{h}_t$ affect *cost*, not *validity*.
- LP rule matches PAC with one source; *routing wins* with two.
- Online stays valid under partial feedback; p trades cost vs. error.

Next step

Current data has only one real cheap source. The interesting case is many sources with *heterogeneous costs*, where a scalar threshold no longer suffices.

Thank you! Questions?